

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s).

$$\begin{bmatrix} 0 & 0 & T_{xz} \\ 0 & 0 & 0 \\ T_{xz} & 0 & 0 \end{bmatrix}$$

$$T_{xz} = \mathbb{T}_{xz}^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & 0 & T_{xz} \\ 0 & 0 & 0 \\ -T_{xz} & 0 & 0 \end{bmatrix}$$

$$T_{xz} = -\mathbb{M}_y^{(1)}$$

* Rank 3 tensor (ss). * Rank 3 tensor (sa). * Rank 3 tensor (a). * Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & T_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & T_{yyzx} & 0 \\ 0 & 0 & 0 & 0 & T_{zzzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{yzxy} \\ T_{xxzx} & T_{yyzx} & T_{zzzx} & 0 & 0 & 0 \\ 0 & 0 & 0 & T_{yzxy} & 0 & 0 \end{bmatrix}$$

$$T_{xxzx} = -\mathbb{T}_{4y}^{\alpha(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} - \mathbb{T}_y^{\beta(1)}$$

$$T_{yyzx} = \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_y^{\beta(1)}$$

$$T_{zzzx} = \mathbb{T}_{4y}^{\alpha(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} - \mathbb{T}_y^{\beta(1)}$$

$$T_{yzxy} = \mathbb{T}_{xz}^{(2)} + 2\mathbb{T}_y^{\beta(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & T_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & T_{yyzx} & 0 \\ 0 & 0 & 0 & 0 & T_{zzzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{yzxy} \\ -T_{xxzx} & -T_{yyzx} & -T_{zzzx} & 0 & 0 & 0 \\ 0 & 0 & 0 & -T_{yzxy} & 0 & 0 \end{bmatrix}$$

$$T_{xxzx} = -2\mathbb{M}_y^{(1)} + 2\mathbb{M}_y^{\alpha(1)} + 2\mathbb{M}_y^{\beta(1)} + \mathbb{T}_{xz}^{(3)}$$

$$T_{yyzx} = -4\mathbb{M}_y^{\beta(1)} + \mathbb{T}_{xz}^{(3)}$$

$$T_{zzzx} = 2\mathbb{M}_y^{(1)} - 2\mathbb{M}_y^{\alpha(1)} + 2\mathbb{M}_y^{\beta(1)} + \mathbb{T}_{xz}^{(3)}$$

$$T_{yzxy} = \mathbb{M}_y^{(1)} + 4\mathbb{M}_y^{\alpha(1)}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & 0 & T_{yzxy} \\ 0 & 0 & 0 \\ T_{yzxy} & 0 & 0 \end{bmatrix}$$

$$T_{yzxy} = 2\mathbb{T}_{xz}^{(4)}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & 0 & T_{yzxy} \\ 0 & 0 & 0 \\ -T_{yzxy} & 0 & 0 \end{bmatrix}$$

$$T_{yzxy} = \mathbb{M}_y^{(2)}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & T_{xxzx} & 0 \\ 0 & T_{yyzx} & 0 \\ 0 & T_{zzzx} & 0 \\ 0 & 0 & T_{yzxy} \\ 0 & 0 & 0 \\ T_{xyyz} & 0 & 0 \end{bmatrix}$$

$$T_{xxzx} = \mathbb{M}_y^{(3)} - \mathbb{M}_y^{\alpha(2)} - \mathbb{M}_y^{\beta(2)} + 2\mathbb{T}_{xz}^{(5)}$$

$$T_{yyzx} = \mathbb{M}_y^{(3)} + 2\mathbb{M}_y^{(4)} + 2\mathbb{M}_y^{\alpha(2)}$$

$$T_{zzzx} = \mathbb{M}_y^{(3)} - \mathbb{M}_y^{\alpha(2)} + \mathbb{M}_y^{\beta(2)} - 2\mathbb{T}_{xz}^{(5)}$$

$$T_{yzxy} = \mathbb{M}_y^{(4)} - \mathbb{M}_y^{\alpha(2)} + \mathbb{M}_y^{\beta(2)} + \mathbb{T}_{xz}^{(5)}$$

$$T_{xyyz} = \mathbb{M}_y^{(4)} - \mathbb{M}_y^{\alpha(2)} - \mathbb{M}_y^{\beta(2)} - \mathbb{T}_{xz}^{(5)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & T_{yzxy} \\ T_{zxxx} & T_{zxyy} & T_{zxzz} & 0 & 0 & 0 \\ 0 & 0 & 0 & T_{xyyz} & 0 & 0 \end{bmatrix}$$

$$T_{yzxy} = \mathbb{M}_y^{(6)} - \mathbb{M}_y^{\alpha(3)} - \mathbb{M}_y^{\beta(3)} - \mathbb{T}_{xz}^{(6)}$$

$$T_{zxxx} = \mathbb{M}_y^{(5)} - \mathbb{M}_y^{\alpha(3)} - \mathbb{M}_y^{\beta(3)} + 2\mathbb{T}_{xz}^{(6)}$$

$$T_{zxyy} = \mathbb{M}_y^{(5)} + 2\mathbb{M}_y^{(6)} + 2\mathbb{M}_y^{\alpha(3)}$$

$$T_{zxzz} = \mathbb{M}_y^{(5)} - \mathbb{M}_y^{\alpha(3)} + \mathbb{M}_y^{\beta(3)} - 2\mathbb{T}_{xz}^{(6)}$$

$$T_{xyyz} = \mathbb{M}_y^{(6)} - \mathbb{M}_y^{\alpha(3)} + \mathbb{M}_y^{\beta(3)} + \mathbb{T}_{xz}^{(6)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & T_{xxzx} & 0 & 0 & T_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & T_{yyzx} & 0 & 0 & T_{yyxz} & 0 \\ 0 & 0 & 0 & 0 & T_{zzzx} & 0 & 0 & T_{zzxz} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{yzxy} & 0 & 0 & T_{yzyx} \\ T_{zxxx} & T_{zxyy} & T_{zxzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & T_{xyyz} & 0 & 0 & T_{xyzy} & 0 & 0 \end{bmatrix}$$

$$T_{xxzx} = -2\mathbb{M}_y^{(1)} + \mathbb{M}_y^{(3)} + 2\mathbb{M}_y^{\alpha(1)} - \mathbb{M}_y^{\alpha(2)} + 2\mathbb{M}_y^{\beta(1)} - \mathbb{M}_y^{\beta(2)} - \mathbb{T}_{4y}^{\alpha(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} + 2\mathbb{T}_{xz}^{(5)} - \mathbb{T}_y^{\beta(1)}$$

$$T_{xxxz} = -2\mathbb{M}_y^{(1)} - \mathbb{M}_y^{(3)} + 2\mathbb{M}_y^{\alpha(1)} + \mathbb{M}_y^{\alpha(2)} + 2\mathbb{M}_y^{\beta(1)} + \mathbb{M}_y^{\beta(2)} - \mathbb{T}_{4y}^{\alpha(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} - 2\mathbb{T}_{xz}^{(5)} - \mathbb{T}_y^{\beta(1)}$$

$$T_{yyzx} = \mathbb{M}_y^{(3)} + 2\mathbb{M}_y^{(4)} + 2\mathbb{M}_y^{\alpha(2)} - 4\mathbb{M}_y^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + \mathbb{T}_{xz}^{(3)} + 2\mathbb{T}_y^{\beta(1)}$$

$$T_{yyxz} = -\mathbb{M}_y^{(3)} - 2\mathbb{M}_y^{(4)} - 2\mathbb{M}_y^{\alpha(2)} - 4\mathbb{M}_y^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + \mathbb{T}_{xz}^{(3)} + 2\mathbb{T}_y^{\beta(1)}$$

$$T_{zzzx} = 2\mathbb{M}_y^{(1)} + \mathbb{M}_y^{(3)} - 2\mathbb{M}_y^{\alpha(1)} - \mathbb{M}_y^{\alpha(2)} + 2\mathbb{M}_y^{\beta(1)} + \mathbb{M}_y^{(2)} + \mathbb{T}_{4y}^{\alpha(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} - 2\mathbb{T}_{xz}^{(5)} - \mathbb{T}_y^{\beta(1)}$$

$$T_{zzxz} = 2\mathbb{M}_y^{(1)} - \mathbb{M}_y^{(3)} - 2\mathbb{M}_y^{\alpha(1)} + \mathbb{M}_y^{\alpha(2)} + 2\mathbb{M}_y^{\beta(1)} - \mathbb{M}_y^{\beta(2)} + \mathbb{T}_{4y}^{\alpha(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} + 2\mathbb{T}_{xz}^{(5)} - \mathbb{T}_y^{\beta(1)}$$

$$T_{yzxy} = \mathbb{M}_y^{(1)} + \mathbb{M}_y^{(4)} + 4\mathbb{M}_y^{\alpha(1)} - \mathbb{M}_y^{\alpha(2)} + \mathbb{M}_y^{\beta(2)} + \mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(5)} + 2\mathbb{T}_y^{\beta(1)}$$

$$T_{yzyx} = \mathbb{M}_y^{(1)} - \mathbb{M}_y^{(4)} + 4\mathbb{M}_y^{\alpha(1)} + \mathbb{M}_y^{\alpha(2)} - \mathbb{M}_y^{\beta(2)} + \mathbb{T}_{xz}^{(2)} - \mathbb{T}_{xz}^{(5)} + 2\mathbb{T}_y^{\beta(1)}$$

$$T_{zxxx} = 2\mathbb{M}_y^{(1)} - 2\mathbb{M}_y^{\alpha(1)} - 2\mathbb{M}_y^{\beta(1)} - \mathbb{T}_{4y}^{\alpha(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} - \mathbb{T}_{xz}^{(3)} - \mathbb{T}_y^{\beta(1)}$$

$$T_{zxyy} = 4\mathbb{M}_y^{\beta(1)} + \mathbb{T}_{xz}^{(1)} - \mathbb{T}_{xz}^{(3)} + 2\mathbb{T}_y^{\beta(1)}$$

$$T_{zxzz} = -2\mathbb{M}_y^{(1)} + 2\mathbb{M}_y^{\alpha(1)} - 2\mathbb{M}_y^{\beta(1)} + \mathbb{T}_{4y}^{\alpha(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} - \mathbb{T}_{xz}^{(3)} - \mathbb{T}_y^{\beta(1)}$$

$$T_{xyyz} = -\mathbb{M}_y^{(1)} + \mathbb{M}_y^{(4)} - 4\mathbb{M}_y^{\alpha(1)} - \mathbb{M}_y^{\alpha(2)} - \mathbb{M}_y^{\beta(2)} + \mathbb{T}_{xz}^{(2)} - \mathbb{T}_{xz}^{(5)} + 2\mathbb{T}_y^{\beta(1)}$$

$$T_{xyzy} = -\mathbb{M}_y^{(1)} - \mathbb{M}_y^{(4)} - 4\mathbb{M}_y^{\alpha(1)} + \mathbb{M}_y^{\alpha(2)} + \mathbb{M}_y^{\beta(2)} + \mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(5)} + 2\mathbb{T}_y^{\beta(1)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 & T_{yzxy} & 0 & 0 & T_{yzyx} \\ T_{zxzx} & T_{zxyy} & T_{zxzz} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & T_{xyyz} & 0 & 0 & T_{xyzy} & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} T_{yzxy} &= \mathbb{M}_y^{(2)} + \mathbb{M}_y^{(6)} - \mathbb{M}_y^{\alpha(3)} - \mathbb{M}_y^{\beta(3)} + 2\mathbb{T}_{xz}^{(4)} - \mathbb{T}_{xz}^{(6)} \\ T_{yzyx} &= -\mathbb{M}_y^{(2)} + \mathbb{M}_y^{(6)} - \mathbb{M}_y^{\alpha(3)} - \mathbb{M}_y^{\beta(3)} - 2\mathbb{T}_{xz}^{(4)} - \mathbb{T}_{xz}^{(6)} \\ T_{zxzx} &= \mathbb{M}_y^{(5)} - \mathbb{M}_y^{\alpha(3)} - \mathbb{M}_y^{\beta(3)} + 2\mathbb{T}_{xz}^{(6)} \\ T_{zxyy} &= \mathbb{M}_y^{(5)} + 2\mathbb{M}_y^{(6)} + 2\mathbb{M}_y^{\alpha(3)} \\ T_{zxzz} &= \mathbb{M}_y^{(5)} - \mathbb{M}_y^{\alpha(3)} + \mathbb{M}_y^{\beta(3)} - 2\mathbb{T}_{xz}^{(6)} \\ T_{xyyz} &= -\mathbb{M}_y^{(2)} + \mathbb{M}_y^{(6)} - \mathbb{M}_y^{\alpha(3)} + \mathbb{M}_y^{\beta(3)} + 2\mathbb{T}_{xz}^{(4)} + \mathbb{T}_{xz}^{(6)} \\ T_{xyzy} &= \mathbb{M}_y^{(2)} + \mathbb{M}_y^{(6)} - \mathbb{M}_y^{\alpha(3)} + \mathbb{M}_y^{\beta(3)} - 2\mathbb{T}_{xz}^{(4)} + \mathbb{T}_{xz}^{(6)} \end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & 0 & T_{xxxx} \\ 0 & 0 & 0 \\ T_{zzzx} & 0 & 0 \\ T_{yyzx} & 0 & 0 \\ 0 & 0 & T_{zzxz} \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ T_{zxzx} & 0 & 0 \\ 0 & 0 & T_{xyyz} \\ 0 & T_{xyyz} + \frac{T_{zzxz}}{2} - \frac{T_{zzzx}}{2} & 0 \end{bmatrix}$$

$$\begin{aligned} T_{xxxx} &= -\mathbb{T}_{4y}^{\alpha(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} - 2\mathbb{T}_{xz}^{(5)} - \mathbb{T}_y^{\beta(1)} \\ T_{zzzx} &= \mathbb{T}_{4y}^{\alpha(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} - 2\mathbb{T}_{xz}^{(5)} - \mathbb{T}_y^{\beta(1)} \\ T_{yyzx} &= \mathbb{T}_{xz}^{(1)} + \mathbb{T}_{xz}^{(3)} + 2\mathbb{T}_y^{\beta(1)} \\ T_{zzzz} &= \mathbb{T}_{4y}^{\alpha(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} + 2\mathbb{T}_{xz}^{(5)} - \mathbb{T}_y^{\beta(1)} \\ T_{zxzx} &= -\mathbb{T}_{4y}^{\alpha(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} - \mathbb{T}_{xz}^{(3)} - \mathbb{T}_y^{\beta(1)} \\ T_{xyyz} &= \mathbb{T}_{xz}^{(2)} - \mathbb{T}_{xz}^{(5)} + 2\mathbb{T}_y^{\beta(1)} \end{aligned}$$